

L05 — Insertion and Deletion in Arrays

Unit: 1 | Chapter: Fundamentals of Data Structures & Arrays | BT Level: BT3 | CO: CO2

Learning Objectives

- Implement insertion at any position in an array
 - Implement deletion from any position in an array
 - Analyze the time complexity of each operation
 - Understand why arrays have $O(n)$ insertion/deletion cost
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1. Recap: Arrays in Memory

Arrays occupy **contiguous memory**. This is great for access but problematic for insertion/deletion — shifting is required to maintain the contiguous property.

Before insertion of 99 at index 2:

Index:	0	1	2	3	4
Value:	[10]	[20]	[30]	[40]	[50]

After insertion:

Index:	0	1	2	3	4	5
Value:	[10]	[20]	[99]	[30]	[40]	[50]
		↑	↑	↑	elements shifted right	

2. Insertion in an Array

2.1 Insertion at the End — $O(1)$

The simplest case: no shifting required.

```
def insert_at_end(arr, size, capacity, element):
    if size == capacity:
        raise Exception("Array is full")
    arr[size] = element
    return size + 1 # new size

# Example
arr = [10, 20, 30, 0, 0] # capacity = 5, size = 3
size = insert_at_end(arr, 3, 5, 40)
# arr = [10, 20, 30, 40, 0], size = 4
```

Time: $O(1)$ | **Space:** $O(1)$

2.2 Insertion at a Given Position — $O(n)$

Must shift all elements from position `pos` to `size-1` one step to the right.

```
def insert_at_position(arr, size, capacity, pos, element):
    """
    Insert element at index pos (0-based).
    Valid pos: 0 to size (inclusive).
    """
    if size == capacity:
        raise Exception("Array is full")
    if pos < 0 or pos > size:
        raise Exception("Invalid position")

    # Shift elements right from size-1 down to pos
    for i in range(size, pos, -1):
        arr[i] = arr[i - 1]

    arr[pos] = element
    return size + 1
```

Step-by-step trace for insert 99 at pos=2 in [10,20,30,40,50]:

Step	Action	Array State
Start	—	[10, 20, 30, 40, 50, _]
i=5	arr[5] = arr[4] = 50	[10, 20, 30, 40, 50, 50]
i=4	arr[4] = arr[3] = 40	[10, 20, 30, 40, 40, 50]
i=3	arr[3] = arr[2] = 30	[10, 20, 30, 30, 40, 50]
i=2	arr[2] = 99	[10, 20, 99, 30, 40, 50]

Time: $O(n)$ worst case (insertion at index 0 shifts all n elements)

Space: $O(1)$

2.3 Insertion at Beginning — $O(n)$

Special case of position insertion with `pos = 0`. All n elements must shift.

```
def insert_at_beginning(arr, size, capacity, element):
    return insert_at_position(arr, size, capacity, 0, element)
```

3. Deletion from an Array

3.1 Deletion from the End — $O(1)$

Simply reduce the logical size (no actual data erasure needed, though you can zero it).

```
def delete_from_end(arr, size):
    if size == 0:
        raise Exception("Array is empty")
    arr[size - 1] = 0 # Optional: clear the slot
    return size - 1 # new size
```

Time: $O(1)$ | **Space:** $O(1)$

3.2 Deletion at a Given Position — $O(n)$

Shift all elements from $\text{pos}+1$ to $\text{size}-1$ one step to the left.

```
def delete_at_position(arr, size, pos):
    """
    Delete element at index pos (0-based).
    Valid pos: 0 to size-1.
    """
    if size == 0:
        raise Exception("Array is empty")
    if pos < 0 or pos >= size:
        raise Exception("Invalid position")

    deleted_element = arr[pos]

    # Shift elements left from pos+1 to size-1
    for i in range(pos, size - 1):
        arr[i] = arr[i + 1]

    arr[size - 1] = 0 # Optional: clear last slot
    return size - 1, deleted_element
```

Step-by-step trace for delete at $\text{pos}=2$ in $[10,20,30,40,50]$:

Step	Action	Array State
Start	—	$[10, 20, 30, 40, 50]$
$i=2$	$\text{arr}[2] = \text{arr}[3] = 40$	$[10, 20, 40, 40, 50]$
$i=3$	$\text{arr}[3] = \text{arr}[4] = 50$	$[10, 20, 40, 50, 50]$
Clear	$\text{arr}[4] = 0$	$[10, 20, 40, 50, 0]$

Time: $O(n)$ worst case (deletion at index 0 shifts all $n-1$ elements)

Space: $O(1)$

3.3 Deletion at Beginning — $O(n)$

```
def delete_from_beginning(arr, size):
    return delete_at_position(arr, size, 0)
```

4. Search Before Deletion

Often we don't know the index — we only know the value. We must first search for it.

```
def delete_by_value(arr, size, target):
    """Delete first occurrence of target."""
    # Step 1: Find the index —  $O(n)$ 
    pos = -1
    for i in range(size):
        if arr[i] == target:
            pos = i
```

```

        break

    if pos == -1:
        return size, False # not found

    # Step 2: Delete at found position - O(n)
    size, _ = delete_at_position(arr, size, pos)
    return size, True

# Total: O(n) + O(n) = O(n)

```

5. Complexity Summary

Operation	Best Case	Worst Case	Average Case
Insert at end	O(1)	O(1)	O(1)
Insert at position i	O(1)	O(n)	O(n)
Insert at beginning	O(n)	O(n)	O(n)
Delete from end	O(1)	O(1)	O(1)
Delete at position i	O(1)	O(n)	O(n)
Delete from beginning	O(n)	O(n)	O(n)

Why O(n)? In the worst case, inserting at or deleting from index 0 requires shifting all n elements.

6. Complete Demo

```

class DynamicArray:
    def __init__(self, capacity):
        self.arr = [0] * capacity
        self.capacity = capacity
        self.size = 0

    def insert(self, pos, val):
        if self.size >= self.capacity:
            raise Exception("Array full")
        for i in range(self.size, pos, -1):
            self.arr[i] = self.arr[i - 1]
        self.arr[pos] = val
        self.size += 1

    def delete(self, pos):
        if pos < 0 or pos >= self.size:
            raise Exception("Invalid index")
        for i in range(pos, self.size - 1):
            self.arr[i] = self.arr[i + 1]
        self.size -= 1

    def display(self):

```

```
print(self.arr[:self.size])
```

```
da = DynamicArray(10)
da.insert(0, 10)      # [10]
da.insert(1, 20)      # [10, 20]
da.insert(2, 30)      # [10, 20, 30]
da.insert(1, 99)      # [10, 99, 20, 30]
da.display()          # → [10, 99, 20, 30]
da.delete(1)          # [10, 20, 30]
da.display()          # → [10, 20, 30]
```

Summary

- **Insertion at end:** $O(1)$ — just append.
 - **Insertion at position i :** $O(n)$ — shift elements i to $n-1$ right by one.
 - **Deletion from end:** $O(1)$ — reduce logical size.
 - **Deletion at position i :** $O(n)$ — shift elements $i+1$ to $n-1$ left by one.
 - The need to maintain contiguity is why array insertion/deletion is costly — this motivates **linked lists**.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 10.1 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 3.2 (Springer, 2020)